

2026 Spring

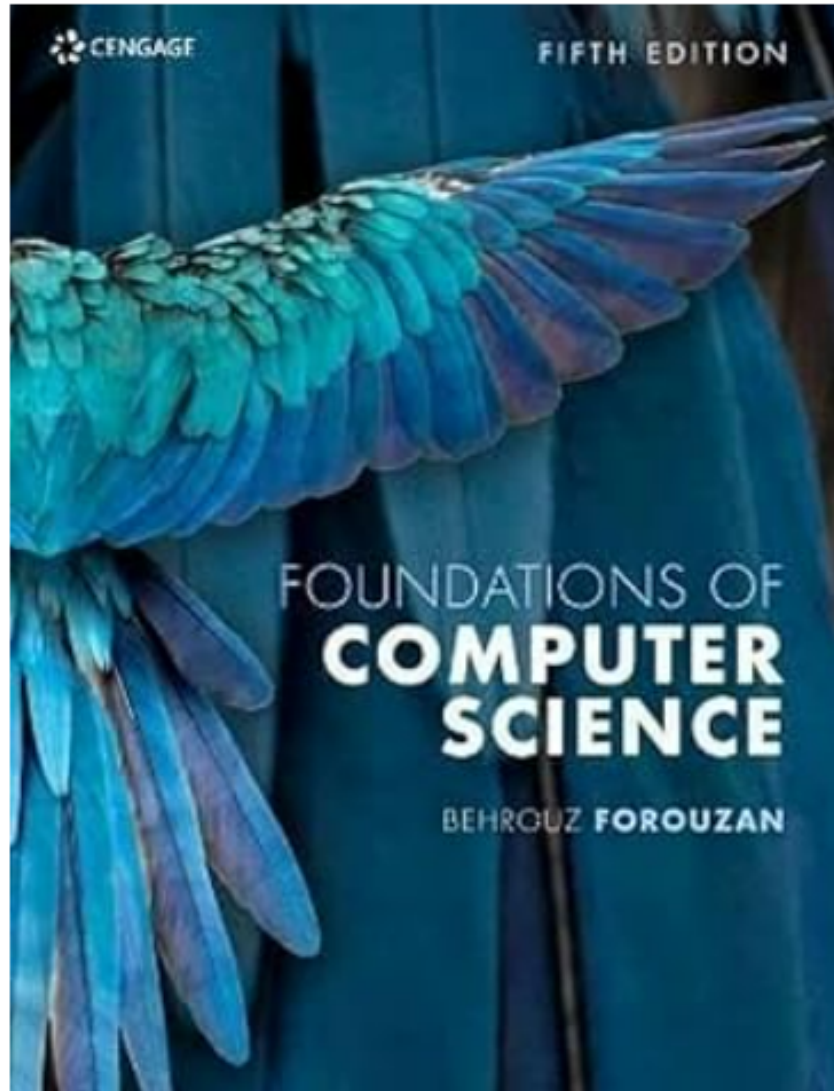
Course ID: 1142-1764

Foundations of Computer Science



14-Databases

The Textbook



Foundations of Computer Science Paperback



作者 [Behrouz Forouzan](#) (Author)

The symposium was held Oct.-Nov. 1989, Research Triangle Park, North Carolina. One hundred papers in theoretical computer science treat dispersers, deterministic amplification, and weak random sources; efficient NC algorithms for set cover with applications to learning and geometry; the inverse of automorphism in polynomial time; and speeding-up linear programming using fast matrix multiplication. Acidic paper; no subject index. Annotation copyright Book News, Inc. Portland, Or.

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English

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Flip Classroom

Padlet Collaborative Learning

Introduction to Computer Science | Modules 5 – 10



M5 Networking



M6 Compute



M7 Storage



M8 Databases



M9 Architecture



M10 Auto Scaling



Your Group Builds the Knowledge Base — Together

Grading

The report's grading includes:

- 20% teacher's score
- 20% group score
- 60% peer evaluation.

Group leaders:

Module 6 - 411425209 LE TUAN ANH 黎俊英

Module 7 - 411416878 CALVIN THEODORIC LAVENCIO 鄭鎧文

Module 8 - 411416886 MANDY JANNYCIA MAHIDIN 蔡美彦

Module 9 - 411416991 Yansen Ladova 許原盛

Module 10 - E11420600 UDAYA SANKAR HARIHARAN 哈里哈蘭



What Is a Flip Classroom?

1 BEFORE Class

Study the module
watch videos,
read materials at home



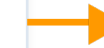
2 PREPARE Padlet

Your group fills in
all 5 sections of
the Padlet template



3 IN Class

Present & discuss
answer classmates'
questions (Q&A)



4 REFLECT

Respond to peer
comments, update
the Padlet



Padlet is the shared workspace where your group documents your learning for each module — visible to the whole class!



Your Padlet Template — 5 Sections

S1



Learning Objectives

State the key learning goals of the module.
What should you understand by the end?

S2



Core Concepts

Summarize essential technical concepts,
terminology, and architectures.

S3



Labs

Document hands-on AWS lab steps,
screenshots, and observations.

S4



Challenges

Record difficult questions, errors
you encountered, and how you solved them.

S5



Q&A & Peer Support

Post questions for the class.
Answer and support your peers.



Tips for a Great Padlet

Add images & diagrams, not just text

Link to AWS documentation & tutorials

Use clear headings in each column

Respond to at least 2 peers in S5

AWS ACADEMY CLOUD FOUNDATIONS

Module 10

Auto Scaling and Monitoring

Elastic Load Balancing · Amazon CloudWatch · Amazon EC2 Auto Scaling

AWS

What You Will Learn

01

Elastic Load Balancing (ELB)

Distribute traffic across multiple targets for high availability

02

Amazon CloudWatch

Monitor AWS resources and applications with metrics, alarms, and logs

03

Amazon EC2 Auto Scaling

Automatically scale EC2 capacity to meet demand

04

Integration & Best Practices

Combine ELB, CloudWatch, and Auto Scaling for resilient architectures

Why Auto Scaling and Monitoring?

The Challenge



Traffic spikes cause server overload



Manual capacity adjustments are slow



Paying for idle resources wastes money



Single points of failure reduce reliability



The AWS Solution



ELB distributes traffic automatically



Auto Scaling adjusts capacity on demand



Pay only for what you use



Multi-AZ deployment for fault tolerance

Elastic Load Balancing (ELB)

Elastic Load Balancing automatically distributes incoming application traffic across multiple targets — such as EC2 instances, containers, and IP addresses — in one or more Availability Zones.

Types of Load Balancers

Application Load Balancer

Layer 7

HTTP/HTTPS routing, content-based routing, microservices

Network Load Balancer

Layer 4

TCP/UDP, ultra-high performance, static IP addresses

Gateway Load Balancer

Layer 3

Deploy & scale virtual network appliances (firewalls, IDS)

How Elastic Load Balancing Works



Amazon CloudWatch

A monitoring and observability service for AWS resources and applications

Metrics

Collect and track key metrics from AWS services automatically. Free tier includes 10 custom metrics.

Alarms

Set thresholds and trigger automated actions when metrics breach defined limits.

Logs

Monitor, store, and access log files from EC2, CloudTrail, Lambda, and more.

Dashboards

Create customizable home pages in the AWS console to monitor resources in one view.

Events

Respond to state changes in AWS resources with automated workflows (EventBridge).

Insights

Query and analyze log data with CloudWatch Logs Insights for fast operational queries.

CloudWatch Metrics & Alarms

EC2	CPUUtilization, NetworkIn/Out, DiskReadOps
ELB	RequestCount, Latency, HTTP Errors (4xx/5xx)
RDS	FreeStorageSpace, DBConnections, ReadLatency
Lambda	Invocations, Duration, Errors, Throttles
S3	BucketSizeBytes, NumberOfObjects, AllRequests



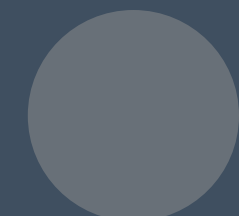
OK

Metric is within the defined threshold



ALARM

Metric breached the threshold — trigger actions



INSUFFICIENT DATA

Not enough data to evaluate the alarm state

When ALARM fires:

- ▶ Send SNS notification (email, SMS, HTTP)
- ▶ Execute Auto Scaling policy
- ▶ Run EC2 action (stop, terminate, recover)

Amazon EC2 Auto Scaling

EC2 Auto Scaling ensures you have the correct number of EC2 instances available to handle your application's load — scaling out to add capacity during demand spikes and scaling in to reduce costs during quiet periods.

Key Benefits

Better Fault Tolerance

Detect and replace unhealthy instances automatically. Distribute across multiple AZs.

Better Availability

Always maintain minimum instance count. Handle unexpected demand increases.

Better Cost Management

Launch instances only when needed. Terminate extras to reduce costs.

Scale Out → Add Instances When Demand ↑

Scale In → Remove Instances When Demand ↓

Auto Scaling Components

1 Launch Template

- AMI (Amazon Machine Image)
- Instance type (e.g., t3.micro)
- Key pair & security groups
- IAM role & user data script
- EBS volume configuration

2 Auto Scaling Group

- Minimum capacity (floor)
- Maximum capacity (ceiling)
- Desired capacity (target)
- VPC subnets & AZs
- Health check settings

3 Scaling Policies

- Target Tracking Scaling
- Step Scaling
- Simple Scaling
- Scheduled Scaling
- Predictive Scaling

Scaling Policies

Target Tracking Scaling

RECOMMENDED

Scales to maintain a specific metric at a target value.

Example: Keep average CPU at 50% → Auto Scaling adds/removes instances automatically.

Step Scaling

FLEXIBLE

Scale in steps based on the size of the alarm breach.

Example: CPU 70-80%: add 1 instance. CPU 80-90%: add 2. CPU >90%: add 4.

Scheduled Scaling

PREDICTABLE

Set min/max capacity on a schedule for predictable traffic patterns.

Example: Every Monday 8 AM: set min=10. Every Monday 8 PM: set min=2.

Predictive Scaling

ML-POWERED

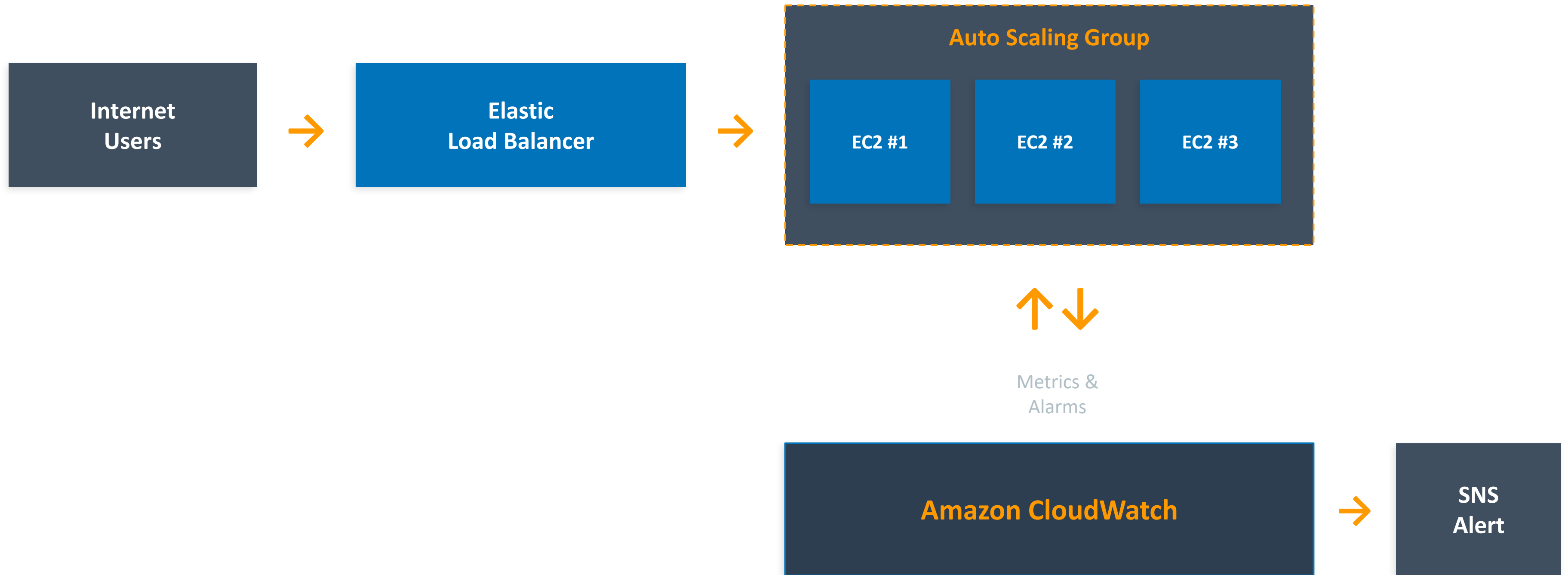
Uses machine learning to forecast future traffic and pre-scale capacity.

Example: Analyzes 14 days of data to predict daily/weekly load patterns.



Cool-down period (default 300s): prevents Auto Scaling from launching/terminating additional instances before previous action takes effect.

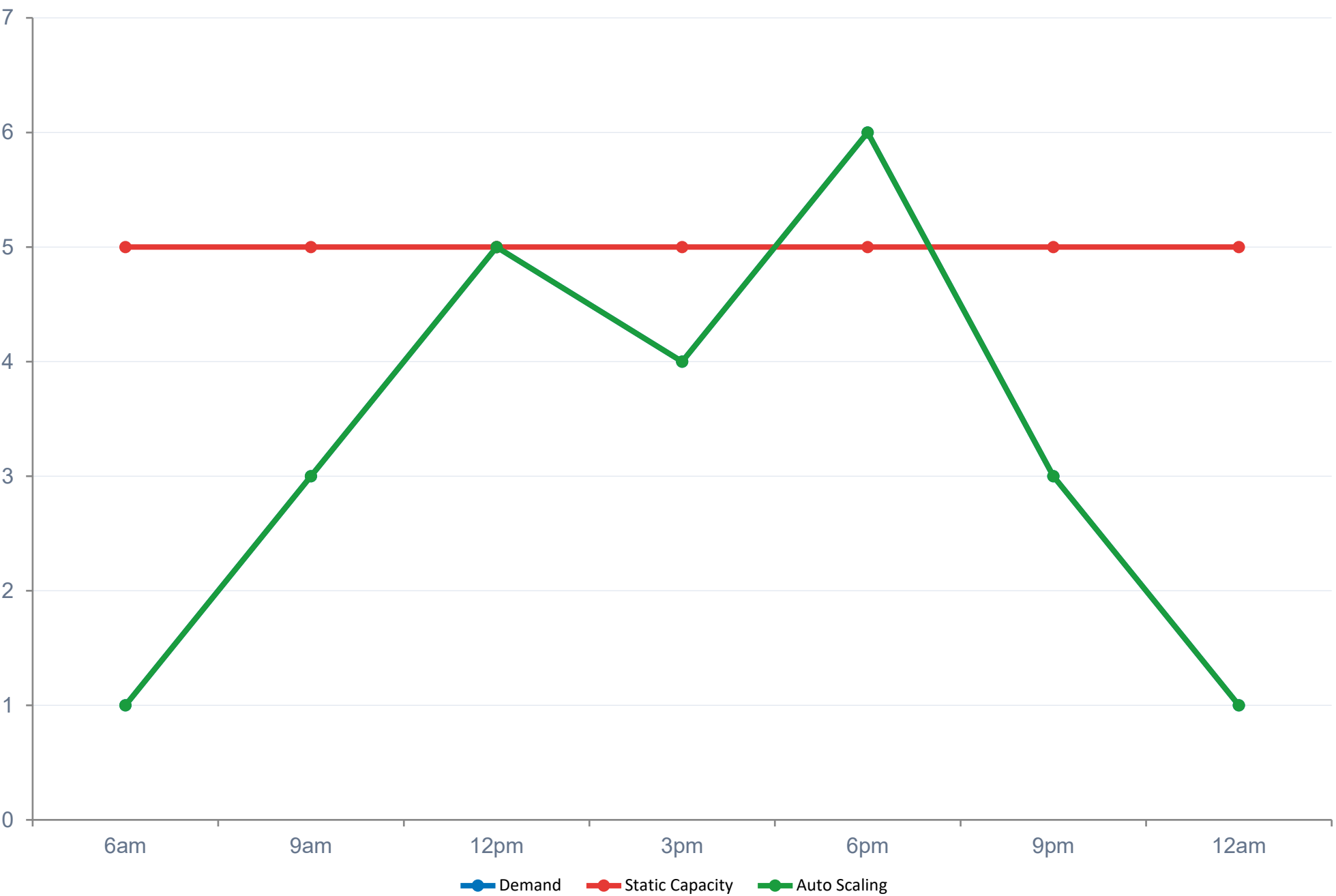
Putting It Together: Integrated Architecture



1. Users send requests through ELB
2. ELB distributes traffic to EC2 instances in the Auto Scaling Group
3. CloudWatch monitors EC2 metrics (CPU, memory, etc.)
4. When metrics breach alarm thresholds, Auto Scaling launches or terminates instances
5. SNS sends notifications to administrators

Cost Optimization with Auto Scaling

Static vs. Dynamic Provisioning



Why Auto Scaling Saves Money

~40%

Cost reduction vs. over-provisioning

0

Idle capacity during low traffic

Auto

Scale to handle unlimited demand

💡 Use Spot Instances in Auto Scaling for up to 90% savings on fault-tolerant workloads!

Knowledge Check

Q1. Which ELB type operates at Layer 7 and supports content-based routing?

A: Application Load Balancer (ALB)

Q2. What are the three states a CloudWatch alarm can be in?

A: OK, ALARM, and INSUFFICIENT_DATA

Q3. Which Auto Scaling policy uses ML to forecast traffic and pre-scale instances?

A: Predictive Scaling

Q4. What is the default cool-down period for Auto Scaling?

A: 300 seconds (5 minutes)

Module 10 Summary



Elastic Load Balancing

Distributes traffic across targets in multiple AZs. ALB (L7), NLB (L4), and GWLB (L3) serve different use cases.



Amazon CloudWatch

Collects metrics from all AWS services, enables alarms with automated actions, and provides centralized logging and dashboards.



EC2 Auto Scaling

Maintains desired capacity with launch templates and scaling policies. Target tracking is the recommended policy type.



Integrated Architecture

Combine ELB + Auto Scaling + CloudWatch to build highly available, fault-tolerant, and cost-efficient applications.

Module 10 Complete!

Auto Scaling and Monitoring

You have learned:

- ✓ How Elastic Load Balancing distributes application traffic for high availability
- ✓ How Amazon CloudWatch provides observability through metrics, alarms, and logs
- ✓ How EC2 Auto Scaling automatically manages compute capacity to match demand
- ✓ How to integrate these services to build resilient, cost-efficient architectures